

Theory of Computation

CSIT — 4th Semester — 2080 — Answer Sheet
Subject Code: CSC262

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Section A

1. Describe the extended transition function of NFA. Construct a NFA, using transition table or transition diagram, over $\Sigma = \{0, 1\}$ that accepts the strings having substring 01 and ends with 1. Show the acceptance of 01111.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. Define CFG. Construct a CFG that generates the language of all palindromes over $\Sigma = \{a, b\}$ that do not contain the substring ab. Show the leftmost derivation and construct the equivalent parse tree for string babbbab.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. How Turing Machine is used as a computing function? Construct a TM for simulating a function $f(x) = 2x$ for $x = \{1\}^+ \cup \{\epsilon\}$. Iterate the TM to compute 111 and generate the output.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section B

Attempt any Eight questions[8×5=40]

4. Differentiate Kleen closure from Positive closure. Compute positive and Kleen closure over $\Sigma = \{a, b\}$.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Design a Mealy machine over $\Sigma = \{1, 0\}$ that generates output 'A' if the input string ends with 11 else output 'B' if the string ends with 00.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Construct regular expression over $\{1 \dots 9\}$ that represents a. strings of even number with length 4 starting with 1 and ending with 8 b. strings starting with odd number and ending with even numbers

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Prove that the language $L = \{a^n b^n c^n \mid n > 0\}$ is not context free.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Construct a PDA that accepts strings over $\Sigma = \{a, b\}$ that contains equal number of a's followed by equal number of b's. Show acceptance of aabb and aaabbb.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. Describe how multi-stack TM is different from the semi-infinite tape TM?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. What is intractability? Define time and space complexity of Turing Machine.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. How conversion of PDA to CFG is done? Illustrate with an example

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. State Arden's theorem. Convert following DFA into its regular expression using Arden theorem.

$$0 \rightarrow q_0 \xrightarrow{1} q_1 \xrightarrow{2} q_2 \xrightarrow{1} q_3 \xrightarrow{2} q_2 \xrightarrow{1} q_1 \xrightarrow{2} q_2$$

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.